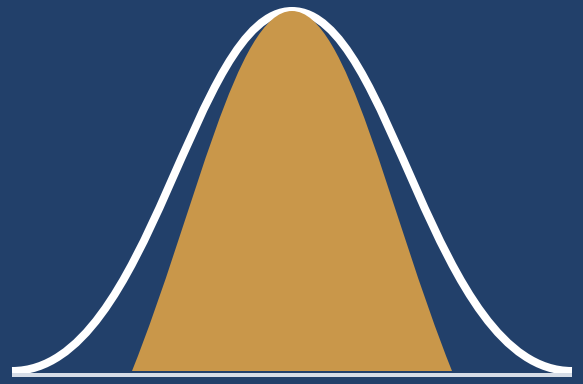
PARENT ROUNDTABLE SERIES

The Why Behind Testing

What a lab test measures, how results are read, and what a test cannot tell you

Deborah M. Burke-Allen, RPh
Registered Pharmacist | Parent education roundtable



START WITH THE QUESTION

Every good test answers a question we have already asked

What are we asking?

Name the symptom or concern first. A test without a question produces numbers without meaning.

What will change?

Decide in advance what a high, low, or normal result would change about the plan.

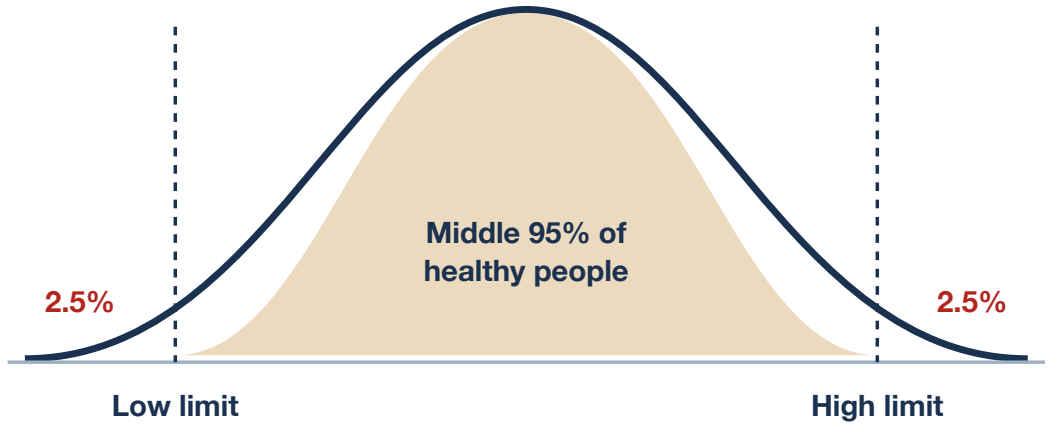
Is now the right time?

Timing, fasting, illness, season, and supplements can all shift a result.

> The best tests are chosen on purpose, not ordered as a panel to see what turns up.

READING THE RANGE

A reference range describes most healthy people, not ideal health



- Labs set the range from the middle 95% of a healthy reference group
- So about 1 in 20 healthy people land outside the range on any single test
- Ranges vary by lab, age, and method, so compare results from the same lab when you can

WORKED EXAMPLE

Reading a vitamin D result: the number, the range, and the next question

Sample result	Value
25-hydroxy vitamin D	**24 ng/mL**
This lab's range	30 - 100 ng/mL
Endocrine Society deficiency	Below 20 ng/mL
Endocrine Society insufficiency	21 - 29 ng/mL

Sample values for teaching. Not a real patient.

How I would read it

- Insufficient, not deficient
- Next questions: season, sun, diet, absorption, and current supplements
- Make one change, then recheck when the clinician advises

Holick MF, et al. J Clin Endocrinol Metab. 2011;96(7):1911-30. PMID 21646368

TRENDS OVER SNAPSHOTS

One result is a snapshot; a trend tells the story



Sample values for teaching. Not a real patient.

- A single value can be caught on a high or low day
- Repeat testing on the same lab and method shows the real direction
- Track results with what changed in between: diet, supplements, season, and symptoms

FUNCTIONAL TESTING

Functional tests add context, so read them carefully

What they can add

- Stool analyses can show digestion markers and patterns of microbial balance
- Organic acids profiles reflect metabolic byproducts from energy production, detoxification, and digestion
- Patterns can guide better questions and conversations

How to read them well

- Validation varies by test and lab
- Read results alongside symptoms, history, and conventional labs
- Treat a pattern as a clue to confirm, not a diagnosis

PHARMACIST LENS

What a test can miss, and what can skew it

- A normal result does not rule out every problem
- An abnormal result is not a diagnosis on its own
- Illness, hydration, fasting, and time of day can shift values

Medicines and supplements matter

High-dose biotin, common in hair, skin, and nail products, can falsely raise or lower some lab results, including thyroid tests and troponin. Tell the lab about every supplement before testing.

TAKE IT WITH YOU

Five questions to ask before and after any test

- **Before:** What question is this test answering?
- **Before:** Do I need to fast or pause any supplements?
- **Before:** Which lab, and will we use the same one next time?
- **After:** How does this compare with my last result?
- **After:** What is the next step, and when do we recheck?

REFERENCES

Sources

1. CLSI EP28-A3c. Defining, establishing, and verifying reference intervals in the clinical laboratory. 2010
2. Holick MF, et al. J Clin Endocrinol Metab. 2011;96(7):1911-30. Evaluation, treatment, and prevention of vitamin D deficiency: an Endocrine Society clinical practice guideline. PMID 21646368
3. U.S. FDA Safety Communication: Biotin may interfere with lab tests. 2017 (updated 2019)